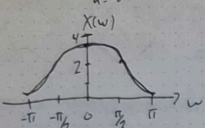


Problem 1, 2, 3: a-b

Assignment 2

1) a) $X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = e^{j\omega} + 2 + e^{-j\omega} = 2 + 2 \cos \omega$



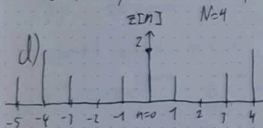
b) $Y(\omega) = \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} = e^{j\omega M} \sum_{n=0}^{2M} e^{-j\omega n} = e^{j\omega M} \left(\frac{1 - e^{-j\omega(2M+1)}}{1 - e^{-j\omega}} \right)$

$$= e^{j\omega M} \left(\frac{e^{-j\omega(2M+1)/2} (e^{j\omega(2M+1)/2} - e^{-j\omega(2M+1)/2})}{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})} \right) \cdot \left(\frac{2j}{2j} \right)$$

$$\Rightarrow Y(\omega) = \frac{\sin(\omega(M + 1/2))}{\sin(\omega/2)}$$

c) They are both discrete time signals which will always have periodic spectrum of real values.

d)
$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} z[n] e^{-j2\pi k n/N}$$



$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n - LN] e^{-j\omega n} \Big|_{\omega = \frac{2\pi k}{N}}$$

$$\Rightarrow c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\omega n} = \frac{1}{N} (2 + 2\cos \omega)$$

e) The sketch would be the same just amplitude will be scaled by $\frac{1}{N}$ or $\frac{1}{40}$. $X(\omega) = N \cdot c_k$

2) a) $x_1[n] \xrightarrow{e^{-j\omega(-1)}} X(\omega)$ Time-shift
 b) $x_2[n] \xrightarrow{X(-\omega)}$ Time-reversal
 c) $x_3[n] \xrightarrow{e^{-j\omega(-1)}} X(-\omega)$ Time-shift and reversal
 d) $x_4[n] \xrightarrow{X^2(\omega)}$ Convolution

3) $y_1[n] = x[n] + 2x[n-1] + x[n-2]$
 a) $y_1(\omega) \xrightarrow{X(\omega) + 2e^{-j\omega}X(\omega) + e^{-2j\omega}X(\omega)} = (1 + 2e^{-j\omega} + e^{-2j\omega})X(\omega)$

$Y(\omega) = H(\omega)X(\omega) \Rightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)}$

$\Rightarrow H_1(\omega) = (1 + 2e^{-j\omega} + e^{-2j\omega}) = (1 + e^{-j\omega})^2 = 4\cos^2(\frac{\omega}{2})e^{-j\omega}$

$Y_2[n] = -0.9Y_1[n-1] + X[n]$
 $Y_2(\omega) = -0.9e^{-j\omega}Y_1(\omega) + X(\omega)$ even function!

$\Rightarrow Y_2(\omega)(1 + 0.9e^{-j\omega}) = X(\omega)$

$\Rightarrow H_2(\omega) = \frac{Y_2(\omega)}{(1 + 0.9e^{-j\omega})Y_1(\omega)} = \frac{1}{(1 + 0.9e^{-j\omega})}$

b) $|H_1(\omega)| = 2 + 2\cos \omega$ & $\phi_1(\omega) = -\omega$

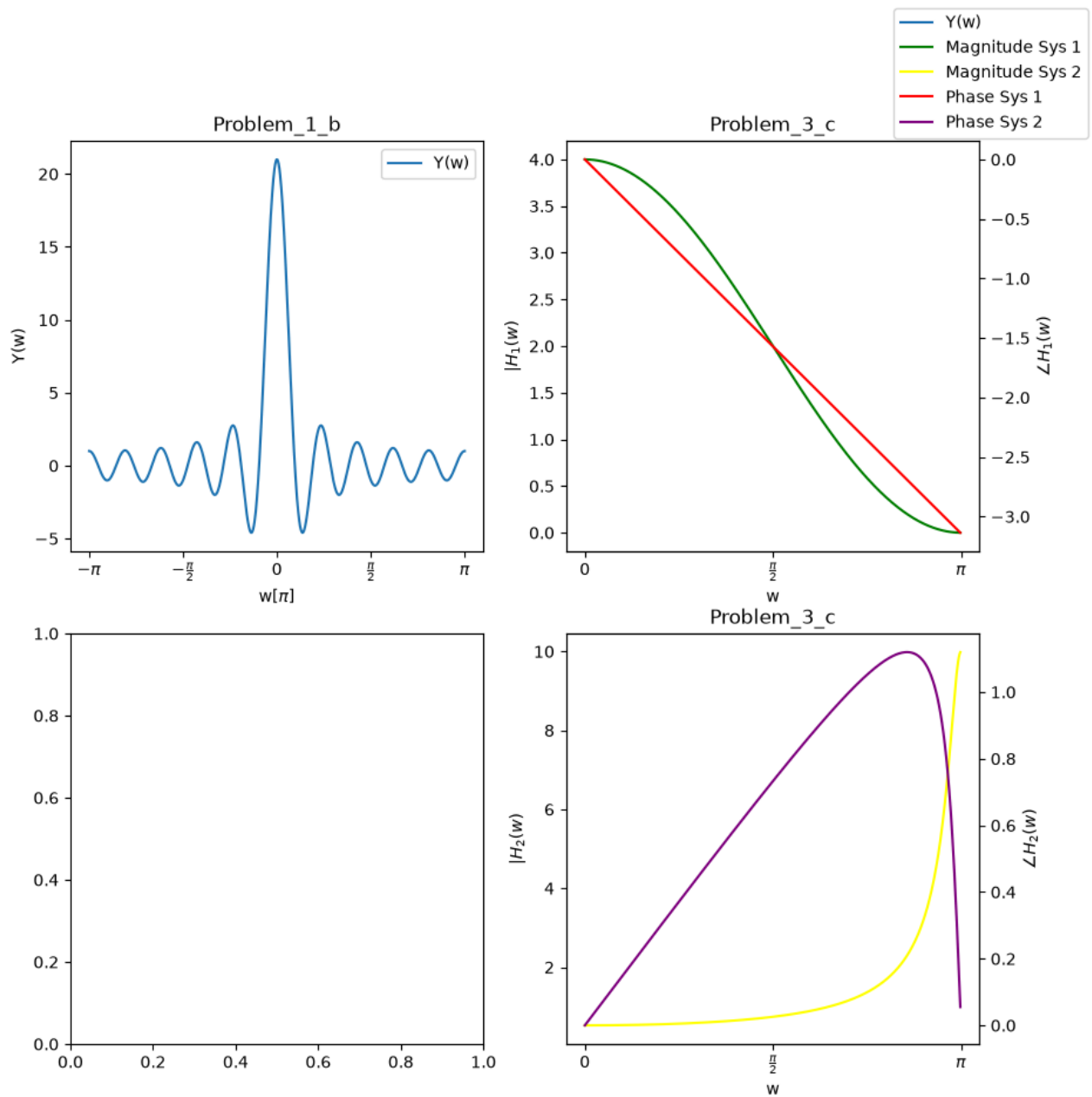
$|H_2(\omega)| = \frac{1}{|1 + 0.9\cos \omega + 0.9j\sin \omega|} = \frac{1}{\sqrt{(1 + 0.9\cos \omega)^2 + 0.9^2 \sin^2 \omega}}$

$\Rightarrow \sqrt{1 + 2 \cdot 0.9 \cos \omega + 0.9^2 \cos^2 \omega + 0.9^2 \sin^2 \omega} = \sqrt{1 + 2 \cdot 0.9 \cos \omega + 2 \cdot 0.9^2 \cos^2 \omega} = \sqrt{1.81 + 1.8 \cos \omega}$

$\phi(\omega) = \tan^{-1} \left(\frac{0.9 \sin \omega}{1 + 0.9 \cos \omega} \right)$ odd func!

Problem 3: c-e (And image 1: b)

c)



d)

It looks like the first system is a lowpass filter as it has a high magnitude for low frequencies and low magnitude for higher frequencies. The second system seems to be a really sharp highpass filter.

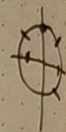
e)

$$3) \quad c) \quad x[n] = \frac{1}{2} \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) \Rightarrow \omega = \frac{\pi}{2}$$

$$\Rightarrow y_1[n] = \frac{1}{2} \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) + \cos\left(\frac{\pi}{2}(n-1) + \frac{\pi}{4}\right) + \frac{1}{2} \cos\left(\frac{\pi}{2}(n-2) + \frac{\pi}{4}\right)$$

$$= \frac{1}{2} \left(\cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) + 2 \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right) + \cos\left(\frac{\pi}{2}n - \frac{3\pi}{4}\right) \right)$$

$$= \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right)$$



$$|y_1[n]| = |\cos\left(\frac{\pi}{2}n - \frac{\pi}{4}\right)| = [-1, 1]$$

$$\phi(\omega) = -\frac{\pi}{4}$$

$$\omega = \frac{\pi}{2} \text{ rad/sample}$$

$$|H_2(\frac{\pi}{2})| = \frac{1}{\sqrt{1^2 + (0,9)^2}} = \frac{1}{\sqrt{1,81}} \Rightarrow |y_2[n]| = \frac{1}{2} \cos\left(\frac{\pi}{2}n + \frac{\pi}{4}\right) \cdot \frac{1}{\sqrt{1,81}}$$

$$= \left[\pm \frac{1}{2 \cdot \sqrt{1,81}} \right]$$

$$\angle H_2(\frac{\pi}{2}) = \tan^{-1}\left(\frac{0,9}{1}\right) \Rightarrow \phi_2(\omega) = \frac{\pi}{4} + \tan^{-1}(0,9)$$

```

import matplotlib.pyplot as plt
import numpy as np
from scipy import signal

M = 10
w = np.linspace(-np.pi, np.pi, 1000)
Y = np.sin(w*(M + 1/2))/(np.sin(w/2))

fig, ax = plt.subplots(2,2, figsize=(10,10))
ax[0,0].plot(w,Y,label="Y(w)")
ax[0,0].set_xlabel(r"$w[\pi]$")
ax[0,0].set_ylabel("Y(w)")
ax[0,0].set_title("Problem_1_b")
ax[0,0].legend()
tick_positions = [-np.pi, -np.pi/2, 0, np.pi/2, np.pi]
tick_labels = [r'$-\pi$', r'$-\frac{\pi}{2}$', r'$0$', r'$\frac{\pi}{2}$', r'$\pi$']
ax[0,0].set_xticks(tick_positions)
ax[0,0].set_xticklabels(tick_labels)

# Problem 3
# sum a_k y_k[n] = sum b_k x_k[n]

a_1 = [1]
b_1 = [1, 2, 1]

w1, h1 = signal.freqz(b_1, a_1)
mag_1 = np.abs(h1)
phase_1 = np.unwrap(np.angle(h1))

ax[0,1].plot(w1, mag_1, label="Magnitude Sys 1", color="green")
ax[0,1].set_xlabel(r"$w$")
ax[0,1].set_ylabel(r"$|H_1(w)$")
tick_positions = [0, np.pi/2, np.pi]
tick_labels = [r'$0$', r'$\frac{\pi}{2}$', r'$\pi$']
ax[0,1].set_xticks(tick_positions)
ax[0,1].set_xticklabels(tick_labels)

ax1_2 = ax[0,1].twinx()
ax1_2.plot(w1, phase_1, label="Phase Sys 1", color="red")
ax1_2.set_ylabel(r"$\angle H_1(w)$")

ax[0,1].set_title("Problem_3_c")

a_2 = [1, 0.9]
b_2 = [1]

```

```

w2, h2 = signal.freqz(b_2, a_2)
mag_2 = np.abs(h2)
phase_2 = np.angle(h2)

ax[1,1].plot(w2, mag_2, label="Magnitude Sys 2", color="yellow")
ax[1,1].set_xlabel(r"w")
ax[1,1].set_ylabel(r"$|H_2(w)$")
tick_positions = [0, np.pi/2, np.pi]
tick_labels = [r'$0$', r'$\frac{\pi}{2}$', r'$\pi$']
ax[1,1].set_xticks(tick_positions)
ax[1,1].set_xticklabels(tick_labels)

ax2_2 = ax[1,1].twinx()
ax2_2.plot(w2, phase_2, label="Phase Sys 2", color="purple")
ax2_2.set_ylabel(r"$\angle H_2(w)$")

ax[1,1].set_title("Problem_3_c")

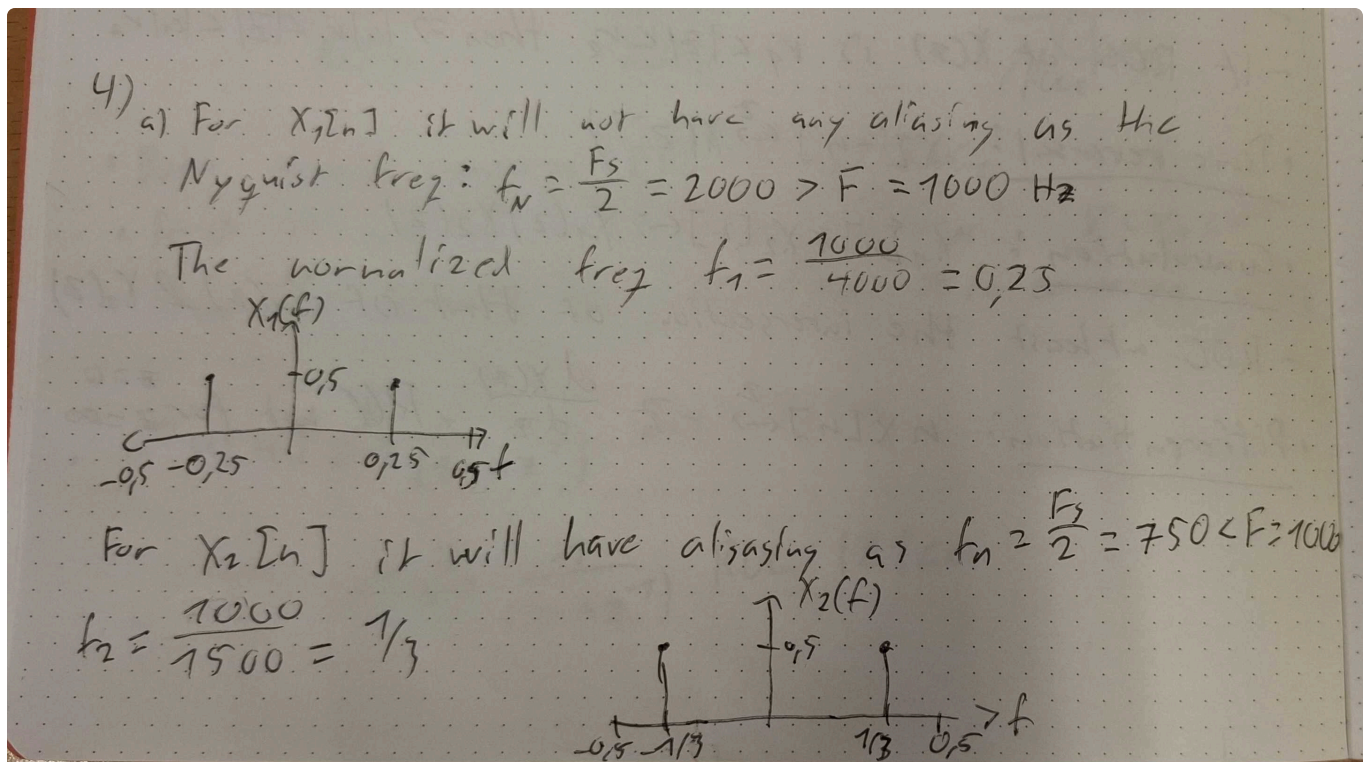
fig.legend()
plt.savefig("Problem_1_b.png")

```



Problem 4:

a)



b)

PYTHON

```
import numpy as np
import sounddevice as sd

# Working script to play tones sampled from a cosine!

F_S1 = 4000
F_S2 = 1500
F = 1000
F_PLAYBACK = 44100

t_total = 1

N1 = np.arange(0, t_total, 1/F_S1)
N2 = np.arange(0, t_total, 1/F_S2)

def x_func(t):
    return np.array(np.cos(2 * np.pi * F * t))

x1 = x_func(N1)
x1_play = np.repeat(x1, int(F_PLAYBACK/F_S1))
x2 = x_func(N2)
x2_play = np.repeat(x2, int(F_PLAYBACK/F_S2))

sd.default.device = 7
sd.play(x2_play, samplerate=F_PLAYBACK)
sd.wait()
```



Because one of them is undersampled so the signal gets aliased. This makes it so it sounds like a 500 Hz sine instead of a 1000 Hz as shown in a).